AMAL A K

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer specialising in scalable SaaS applications using MERN and PERN stacks. Experienced in designing multi-tenant architectures, secure REST APIs, and cloud-based systems. Recently built DiagnomIQ — a healthcare SaaS with Database-per-Tenant isolation, automated Cloudflare DNS provisioning, JWT-based RBAC, and AWS S3 security using the PERN stack.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript

Frontend: React.js, Next.js, Redux, Redux Toolkit, Tailwind CSS, Shadcn/UI, Material UI, React Bootstrap, Recharts, Vite

Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, Middleware Architecture

Databases & ORMS: PostgreSQL, MongoDB, NeonDB, Prisma, Sequelize, Mongoose

Cloud: AWS S3, Amazon Bedrock, Cloudflare DNS, Vercel, Netlify, Render, Docker (basic), CI/CD

Engineering: Multi-Tenant Architecture, System Design, Microservices, Performance Optimisation, Agile

Tools: GitHub, Postman, Antigravity AI, Cursor AI, Windsurf AI, VS Code

PROFESSIONAL EXPERIENCE

Software Engineer Trainee | *Softnotions Technologies Private Limited, Technopark* — Jan 2026 – Apr 2026

- **SaaS Migration:** Led the migration of legacy healthcare systems into a scalable multi-tenant SaaS platform (DiagnomIQ) using the PERN stack.
- **DB Isolation:** Designed Database-per-Tenant architecture with PostgreSQL and Sequelize ensuring 100% tenant data isolation and security.
- **Auto Provisioning:** Built automated tenant onboarding — dynamic database creation, Cloudflare DNS subdomain provisioning, and AWS S3 folder isolation per tenant.
- **RBAC & Middleware:** Implemented JWT-based authentication with Role-Based Access Control and custom Tenant DB Middleware for dynamic request routing.
- **Platform DB:** Architected a centralised Platform Database managing tenant registrations, subscriptions, and system-level operations.
- **AI Integration:** Collaborated with the DevOps team to configure Amazon Bedrock with RAG implementation; created a knowledge base and tested the AI model via the Bedrock API.

MERN Stack Developer | *Relaxplzz Technologies, Technopark* — Apr 2025 – Aug 2025

- Developed scalable frontend features for GiftoMart using React, TypeScript, and Tailwind CSS; built RESTful APIs with Node.js and Express.js.
- Created reusable TypeScript components improving code maintainability; collaborated with product, design, and QA in Agile sprints.

MERN Stack Developer | *SciPy Technologies, Trivandrum* — Jun 2023 – Jan 2024

- Contributed to a microservices-based ERP for an educational institute — RESTful APIs in Node.js, responsive React UIs, and Redux state management.
- Managed MongoDB storage, integrated external APIs via Axios, and validated endpoints with Postman; participated in Agile sprint planning.

PROJECTS

DiagnomIQ – Multi-Tenant Healthcare SaaS | *React · TypeScript · Node.js · PostgreSQL · Sequelize · AWS S3 · Cloudflare · JWT · Amazon Bedrock · Recharts*

- Integrated Amazon Bedrock with RAG — built a medical knowledge base and connected the AI model to the platform via the Bedrock API.
- Built survival probability charts for IPI and aaIPI scoring using Recharts, providing oncologists with visual risk stratification for lymphoma patients.
- Implemented S3 tenant-based folder isolation for sensitive medical data with JWT auth validated against tenant-specific user models.
- Built fully automated tenant provisioning — dynamic database creation and Cloudflare subdomain configuration per tenant.

FinSight – Personal Finance Tracker | *Next.js · TypeScript · PostgreSQL · Prisma ORM · Tailwind CSS · Vercel*

- Full-stack income/expense tracking app with type-safe database access via Prisma ORM and PostgreSQL.
- Built interactive dashboards with category-based analytics and dynamic filtering using shadcn/ui.
- Deployed globally on Vercel with edge optimisation for low-latency performance.

GiftoMart – Gift Management Platform | *React · Node.js · Express.js · MongoDB · TypeScript*

- Built gift creation and tracking modules with secure authentication and reusable TypeScript components.
- Developed REST APIs for product and user services with scalable backend architecture using Node.js, MongoDB, and Mongoose ODM.
- Improved platform stability through debugging and performance optimisation across frontend and backend layers.

ERP System – Educational Institute | *React.js · Node.js · Express.js · MongoDB · Redux · Axios*

- Contributed to a modular ERP supporting academic workflows — responsive React UIs and Node.js backend APIs.
- Managed MongoDB data storage using Mongoose for schema validation and integrated external services using Axios.
- Implemented centralized state management with Redux and validated APIs with Postman; collaborated in Agile sprints.

EDUCATION

Bachelor of Science in Computer Science | Emmanuel College, Vazhichal — 2018 – 2021